Exact Lifting of Coarse Histories in Periodic Upwind Finite Volumes

Antonis Polemitis[1], Nicholas Christakis[2,*], and Dimitris Drikakis[2]

[1]UNIC Evolve, University of Nicosia, Nicosia CY-2417, Cyprus
[2]Institute for Advanced Modelling and Simulation, University of Nicosia, Nicosia CY-2417, Cyprus
[*]Corresponding author: christakis.n@unic.ac.cy

October 12, 2026

Abstract

A prescribed coarse finite-volume history is not automatically compatible with a fine-grid realization and a declared fine-grid update. This short note studies the inverse, or image-space, counterpart of finite-horizon predictive memory. Instead of asking which coarse history is observed from a given fine state, we ask which parent-average histories can be lifted to a fine-grid initial state.

For $P \geq 2$ parent cells and periodic scalar advection discretized by first-order positive upwind finite volumes, we give a complete constructive characterization. For an L-step history, every mass-conserving parent trajectory is realizable if and only if the number of fine children per parent satisfies $r \geq L + 1$. We construct a normalized linear right inverse of the observation map, canonical after fixing the zero-mean flux-gauge and zero-interior convention. We also describe all fine-grid lifts of every realizable short-horizon history, identifying the remaining nonuniqueness as periodic flux-gauge freedom and unexposed interior child values. When $r < L + 1$, arbitrary mass-conserving histories are no longer realizable: admissible histories satisfy an explicit lag-r parent-level recurrence induced by the fact that r fine-grid shifts equal one parent-level shift.

All statements are algebraic over unrestricted real-valued fine states; no positivity, boundedness, invariant-domain, or entropy constraints are imposed. Exact rational certificates reproduce a normalized lift, the long-horizon recurrence, and an explicit nonnegative mass-conserving but nonrealizable history. The results provide a sharp discrete L-step minimal child-count law for the declared finite-volume hierarchy.

Keywords: finite-volume methods; coarse graining; trajectory realizability; upwind advection; behavioral systems; minimal child count.

Mathematics Subject Classification (2020): 65M08; 93B07; 93B03; 93B25.

1 Introduction

Finite-volume methods evolve cell averages through numerical fluxes at cell interfaces [1, 2]. In coarse/fine coupling, adaptive mesh refinement, multirate integration, reduced modelling, and learned discretization, one may ask whether a proposed coarse trajectory is compatible with a declared fine-grid computation [3–9].

The companion finite-volume erasure paper studies the forward observation problem for a periodic scalar upwind hierarchy [10]. It asks: given a fine initial state, what parent-average history is observed, how much hidden memory is needed to predict that history, how well conditioned is the corresponding information, and when is it dissipated? A second companion note studies one-step information boundaries [11]: it shows that parent averages together with

complete initial parent-interface trace pairs close one forward-Euler transition but do not generally close a completed second-order strong-stability-preserving Runge–Kutta, equivalently Heun, transition, and it shows that conservative one-step closure does not determine entropy-accounting closure.

The present note asks the image-space counterpart: given a parent-average history, when does some fine-grid initial state realize it? The companion forward-analysis paper determines rank, nullspace, conditioning, and predictive memory of the fine-state-to-history map. This note solves the inverse problem for the same declared hierarchy: it constructs a normalized right inverse, describes every fiber over an admissible history, and gives a necessary-and-sufficient description of the image.

This inverse viewpoint is closely related to behavioral systems theory, where a dynamical system is studied through the set of trajectories that it generates rather than through one preferred state-space representation [12–14]. It is also related to classical observability, realization, and model reduction [15–21]. The finite-volume-specific contribution here is an exact lifting construction: the recurrence is induced explicitly by r fine-grid shifts producing one parent-level translation, and every realizable short-horizon history has a constructive fine-grid lift with a complete fiber description.

The word lifting is also standard in equation-free and multiscale computation, where a coarse state is mapped to one or more compatible microscopic states before short microscopic simulation is used to advance macroscopic observables [22]. The present use is more specific: it is not merely state lifting at one time, but exact finite-horizon trajectory lifting for a declared finite-volume update, including a complete affine-fiber description.

The result is sharp for $P \geq 2$. Let P be the number of parent cells, r the number of fine children per parent, and L the number of requested update steps. Then:

- if $r \geq L+1$, every parent history through time L that conserves the sum of parent averages is realizable by an unrestricted fine state;
- if $r < L+1$, mass conservation alone is insufficient, and admissible histories satisfy an explicit lag-r recurrence.

Thus

$$r_{\min} = L + 1$$

is a discrete L-step minimal child-count law. It is not a fixed-physical-time mesh-resolution law: if parent widths are held fixed while r changes, then the fine-cell width and, at fixed Courant number, the time step also change.

Contributions. This note makes three contributions:

1. It constructs a normalized linear right inverse that lifts every mass-conserving L-step parent history to a fine-grid initial state when $r \geq L+1$, after fixing the zero-mean flux-gauge and zero-interior convention.
2. It parameterizes all such lifts and determines the exact fiber dimension, separating periodic flux-gauge freedom from unexposed interior child degrees of freedom.
3. It characterizes longer admissible histories exactly by a sharp lag-r recurrence and derives $r_{\min} = L+1$ as a consequence of the image characterization.

All realizability statements are algebraic statements over the complete state space $\mathbb{R}^{Pr}$. They do not assert realizability under positivity, maximum-principle, invariant-domain, entropy, or physical admissibility constraints.

Organization. Section 2 defines the periodic upwind hierarchy and its history map. Section 3 proves the short-horizon image-and-fiber theorem and gives an explicit canonical lift. Section 4 characterizes longer realizable histories by recurrence and proves the sharp minimal child-count law. Section 5 records exact rational certificate examples. Section 6 discusses scope and limitations, and Section 7 concludes.

2 Periodic Upwind Hierarchy and History Map

Consider scalar advection

$$u_t + a u_x = 0, \qquad a > 0,$$

on a periodic one-dimensional domain. The first-order positive upwind finite-volume update is

$$u_i^{n+1} = (1-\lambda)u_i^n + \lambda u_{i-1}^n, \qquad 0 < \lambda \le 1,$$

with periodic indexing, where

$$\lambda = \frac{a\Delta t}{h}$$

is the Courant number.

Let $P \ge 2$ be the number of parent cells, each containing $r \ge 1$ fine children. Then

$$N = Pr$$

is the number of fine cells. Write

$$x = (x_{K,j}), \qquad K = 0, \dots, P-1, \qquad j = 0, \dots, r-1,$$

with parent indices interpreted modulo P.

The parent-average restriction is

$$(Rx)_K = \frac{1}{r}\sum_{j=0}^{r-1} x_{K,j}.$$

Let S be the downstream cyclic shift,

$$(Sx)_{K,j} = \begin{cases} x_{K,j-1}, & j \ge 1, \\ x_{K-1,r-1}, & j = 0, \end{cases}$$

and set

$$A_\lambda = (1-\lambda)I + \lambda S.$$

For a horizon $L \ge 0$, define the history map

$$\mathcal{O}_{L,r} = \begin{bmatrix} R \\ RA_\lambda \\ \vdots \\ RA_\lambda^L \end{bmatrix} : \mathbb{R}^{Pr} \to (\mathbb{R}^P)^{L+1}.$$

Hence

$$\mathcal{O}_{L,r}x = (y^0, y^1, \dots, y^L), \qquad y^n = RA_\lambda^n x.$$

The dependence of $\mathcal{O}_{L,r}$ and of the lifting maps below on the fixed Courant number λ is suppressed in the notation.

Let $\Pi_P : \mathbb{R}^P \to \mathbb{R}^P$ denote the parent cyclic shift,

$$(\Pi_P y)_K = y_{K-1},$$

and define the periodic difference operator

$$B = I - \Pi_P, \qquad (B\Phi)_K = \Phi_K - \Phi_{K-1}.$$

Let

$$\mathbb{R}^P_0 = \left\{ z \in \mathbb{R}^P : \mathbf{1}_P^\top z = 0 \right\}.$$

The operator B maps $\mathbb{R}^P$ into $\mathbb{R}^P_0$, and its kernel on $\mathbb{R}^P$ is the span of $\mathbf{1}_P$. Its restriction to the zero-mean subspace,

$$B_0 := B|_{\mathbb{R}^P_0} : \mathbb{R}^P_0 \to \mathbb{R}^P_0,$$

is invertible. We denote the inverse by B_0^{-1}. Thus $B_0^{-1} g$ is the unique zero-mean register Φ satisfying

$$B\Phi = g$$

for a given $g \in \mathbb{R}^P_0$. Indeed, $B\Phi = 0$ implies that Φ is constant, and the only constant vector in $\mathbb{R}^P_0$ is zero.

A concrete formula is obtained as follows. Set

$$\widehat{\Phi}_0 = 0, \qquad \widehat{\Phi}_K = \sum_{m=1}^{K} g_m, \qquad K = 1, \dots, P-1.$$

Then the zero-mean solution is

$$B_0^{-1} g = \widehat{\Phi} - \frac{1}{P} \left(\sum_{K=0}^{P-1} \widehat{\Phi}_K \right) \mathbf{1}_P.$$

Definition 1 (Mass-conserving parent histories)**.** *For $L \geq 0$, define*

$$\mathcal{M}_{P,L} = \left\{ (y^0, \dots, y^L) \in (\mathbb{R}^P)^{L+1} : \mathbf{1}_P^\top y^n = \mathbf{1}_P^\top y^0 \text{ for } n = 1, \dots, L \right\}.$$

Every history in $\operatorname{im} \mathcal{O}_{L,r}$ lies in $\mathcal{M}_{P,L}$, since the upwind update conserves the fine-grid sum.

3 Short-Horizon Image, Fibers, and Canonical Lifting

Assume

$$0 \leq L \leq r - 1.$$

For $L \geq 1$, define right-collar variables

$$c_{K,j} = x_{K,r-1-j}, \qquad j = 0, \dots, L-1.$$

The outgoing upwind flux through the right face of parent K at time t is

$$\Phi_{K,t} = \lambda (A_\lambda^t x)_{K,r-1} = \lambda \sum_{j=0}^{t} \binom{t}{j} (1-\lambda)^{t-j} \lambda^j c_{K,j}, \qquad t = 0, \dots, L-1.$$

Write this map as

$$\begin{bmatrix} \Phi_{K,0} \\ \Phi_{K,1} \\ \vdots \\ \Phi_{K,L-1} \end{bmatrix} = T_{L,\lambda} \begin{bmatrix} c_{K,0} \\ c_{K,1} \\ \vdots \\ c_{K,L-1} \end{bmatrix}.$$

The matrix $T_{L,\lambda}$ is lower triangular with diagonal entries

$$\lambda, \lambda^2, \dots, \lambda^L,$$

and hence is invertible for $0 < \lambda \le 1$.

The inverse is explicit. If

$$\phi_K = (\Phi_{K,0}, \dots, \Phi_{K,L-1})^\top, \qquad c_K = T_{L,\lambda}^{-1} \phi_K,$$

then for $t = 0, \dots, L-1$,

$$c_{K,t} = \lambda^{-(t+1)} \sum_{j=0}^{t} \binom{t}{j} \big[-(1-\lambda)\big]^{t-j} \Phi_{K,j}, \qquad t = 0, \dots, L-1.$$

This is the binomial inverse of the lower-triangular collar-to-flux map.

Theorem 1 (Short-horizon image, fibers, and canonical lift)**.** *Let $P \ge 2$, $r \ge 1$, $0 < \lambda \le 1$, and $0 \le L \le r-1$. Over the unrestricted fine state space $\mathbb{R}^{Pr}$,*

$$\operatorname{im} \mathcal{O}_{L,r} = \mathcal{M}_{P,L}.$$

Moreover:

1. *There exists a normalized linear right inverse, canonical relative to the zero-mean flux-gauge and zero-interior convention,*
$$\mathcal{L}_{L,r} : \mathcal{M}_{P,L} \to \mathbb{R}^{Pr}$$
such that
$$\mathcal{O}_{L,r} \mathcal{L}_{L,r} = I_{\mathcal{M}_{P,L}}.$$

2. *For every $Y \in \mathcal{M}_{P,L}$, the fiber*
$$\mathcal{O}_{L,r}^{-1}(Y)$$
is a nonempty affine space of dimension
$$P(r-L-1) + L.$$

3. *The fiber degrees of freedom consist of L periodic flux-gauge constants, one for each transition, and*
$$P(r-L-1)$$
unexposed interior child values.

Proof. For $L = 0$, the statement is immediate: every

$$y^0 \in \mathbb{R}^P$$

is lifted canonically by setting

$$x_{K,0} = r y_K^0, \qquad x_{K,j} = 0 \quad (j = 1, \dots, r-1).$$

The fiber dimension is $P(r-1)$, as stated.

Now let $L \ge 1$. The inclusion

$$\operatorname{im} \mathcal{O}_{L,r} \subseteq \mathcal{M}_{P,L}$$

follows from conservation.

Conversely, let

$$Y = (y^0, \dots, y^L) \in \mathcal{M}_{P,L}.$$

For each transition $t = 0, \dots, L-1$, define

$$g^t = r(y^t - y^{t+1}).$$

Since Y conserves the sum of parent averages,

$$\mathbf{1}_P^\top g^t = 0.$$

Choose the unique zero-mean flux register

$$\Phi^t = B_0^{-1} g^t \in \mathbb{R}_0^P.$$

Then

$$B\Phi^t = r(y^t - y^{t+1}),$$

or equivalently

$$y^{t+1} = y^t - \frac{1}{r} B\Phi^t.$$

For each parent K, define the flux history vector

$$\phi_K = (\Phi_{K,0}, \dots, \Phi_{K,L-1})^\top.$$

Set

$$c_K = T_{L,\lambda}^{-1} \phi_K,$$

and assign

$$x_{K,r-1-j} = c_{K,j}, \qquad j = 0, \dots, L-1.$$

Assign all unexposed interior children

$$x_{K,j} = 0, \qquad j = 1, \dots, r-L-1,$$

and finally set

$$x_{K,0} = r y_K^0 - \sum_{j=1}^{r-1} x_{K,j}.$$

This defines the normalized lift $\mathcal{L}_{L,r} Y$. Its parent averages equal y^0, and its outgoing fluxes equal Φ^t for $t = 0, \dots, L-1$. The parent-update identity above then gives

$$\mathcal{O}_{L,r} \mathcal{L}_{L,r} Y = Y.$$

It remains to characterize the fibers. Let $d \in \ker \mathcal{O}_{L,r}$. Then

$$Rd = 0$$

and the parent update shows that each outgoing-flux vector associated with d must satisfy

$$B\Phi^t(d) = 0, \qquad t = 0, \dots, L-1.$$

Hence

$$\Phi^t(d) = \gamma_t \mathbf{1}_P$$

for some independent scalar gauge constants

$$\gamma_0, \dots, \gamma_{L-1}.$$

Inverting $T_{L,\lambda}$, the corresponding collar perturbations are identical in every parent and are uniquely determined by

$$T_{L,\lambda}^{-1} (\gamma_0, \dots, \gamma_{L-1})^\top.$$

The unexposed interior children

$$d_{K,j}, \qquad K = 0, \dots, P-1, \quad j = 1, \dots, r-L-1,$$

are arbitrary. The values $d_{K,0}$ are then uniquely fixed by the conditions $Rd = 0$. Thus the kernel is parameterized by L gauge constants and $P(r - L - 1)$ free interior values, giving dimension

$$L + P(r - L - 1).$$

Every fiber is therefore the translate

$$\mathcal{O}_{L,r}^{-1}(Y) = \mathcal{L}_{L,r}Y + \ker \mathcal{O}_{L,r}.$$

□

Example 1 (Normalized short-horizon lift)**.** *Let*

$$P = 2, \qquad L = 1, \qquad r = 2, \qquad \lambda = \frac{1}{2}.$$

Consider the mass-conserving one-step parent history

$$y^0 = (0, 0), \qquad y^1 = (1, -1).$$

Then

$$g^0 = r(y^0 - y^1) = (-2, 2).$$

The unique zero-mean interface register satisfying

$$B\Phi^0 = g^0$$

is

$$\Phi^0 = (-1, 1).$$

Since

$$\Phi_{K,0} = \frac{1}{2}c_{K,0},$$

the right-collar values are

$$c_{0,0} = -2, \qquad c_{1,0} = 2.$$

The normalized lift is

$$x = (2, -2 \mid -2, 2).$$

Direct calculation gives

$$Rx = (0, 0), \qquad RA_{1/2}x = (1, -1).$$

Thus

$$\mathcal{O}_{1,2}x = (y^0, y^1).$$

At the sharp child count

$$r = L + 1,$$

the fiber dimension in Theorem 1 reduces to

$$L.$$

Thus the only nonuniqueness in a short-horizon lift is the L-dimensional periodic flux gauge.

Remark 1 (Algebraic lifting versus stable lifting)**.** *The theorem establishes exact algebraic surjectivity. It does not assert uniform boundedness or numerical stability of the right inverse. The explicit binomial-inverse formula above contains inverse powers of λ, so prescribed $O(1)$ changes in parent histories may require large collar values when λ is small. The inverse B_0^{-1} also reflects the conditioning of the periodic difference operator and depends on P. Thus exact realizability and stable or physically moderate realizability are distinct questions.*

4 Long-Horizon Realizability

When $L \geq r$, mass conservation is no longer the only constraint. The parent history must satisfy a linear recurrence induced by the identity

$$\big(A_\lambda - (1-\lambda)I\big)^r = \lambda^r S^r.$$

Since

$$RS^r = \Pi_P R,$$

this identity closes at the parent level after r fine-grid shifts.

Define the all-horizon invisible subspace

$$\mathcal{N}_r = \left\{ x \in \mathbb{R}^{Pr} : x_{K,j} = c_j \text{ for all } K, \quad \sum_{j=0}^{r-1} c_j = 0 \right\}.$$

It has dimension $r-1$.

Theorem 2 (Long-horizon image and fibers)**.** *Let $P \geq 2$, $r \geq 1$, $0 < \lambda \leq 1$, and $L \geq r-1$. A parent history*

$$Y = (y^0, \ldots, y^L) \in (\mathbb{R}^P)^{L+1}$$

belongs to $\operatorname{im} \mathcal{O}_{L,r}$ *if and only if:*

1. *the first r parent states have equal sums,*

$$\mathbf{1}_P^\top y^0 = \cdots = \mathbf{1}_P^\top y^{r-1};$$

2. *if $L \geq r$, then for every $n = 0, \ldots, L-r$,*

$$y^{n+r} = \lambda^r \Pi_P y^n - \sum_{j=0}^{r-1} \binom{r}{j} \big[-(1-\lambda)\big]^{r-j} y^{n+j}. \tag{1}$$

For every realizable history,

$$\mathcal{O}_{L,r}^{-1}(Y) = x_\star + \mathcal{N}_r$$

for any one lift $x_\star$, and hence

$$\dim \mathcal{O}_{L,r}^{-1}(Y) = r-1.$$

Proof. Suppose first that

$$Y = \mathcal{O}_{L,r} x.$$

Conservation gives the equality of parent sums. Expanding

$$\big(A_\lambda - (1-\lambda)I\big)^r = \lambda^r S^r$$

gives

$$\sum_{j=0}^{r} \binom{r}{j} \big[-(1-\lambda)\big]^{r-j} A_\lambda^j = \lambda^r S^r.$$

Multiplying by RA_λ^n and using

$$RS^r = \Pi_P R$$

gives

$$\sum_{j=0}^{r} \binom{r}{j} \big[-(1-\lambda)\big]^{r-j} y^{n+j} = \lambda^r \Pi_P y^n.$$

Solving for y^{n+r} gives (1).

Conversely, suppose the stated sum condition and recurrence hold. By Theorem 1, applied at horizon $r-1$, there exists a fine state $x_\star \in \mathbb{R}^{Pr}$ whose first r parent states are

$$y^0, \dots, y^{r-1}.$$

The parent history generated by $x_\star$ satisfies (1). Since it agrees with Y for the first r states, induction gives agreement through time L.

It remains to identify the fiber. Because the temporal transformation from

$$(R, RS, \dots, RS^{r-1})$$

to

$$(R, RA_\lambda, \dots, RA_\lambda^{r-1})$$

is triangular with nonzero diagonal entries, a vector

$$d \in \ker \mathcal{O}_{r-1,r}$$

satisfies

$$RS^t d = 0, \qquad t = 0, \dots, r-1.$$

For $t = 0, \dots, r-2$,

$$r(RS^{t+1}d - RS^t d)_K = d_{K-1,r-1-t} - d_{K,r-1-t}.$$

Thus every child layer

$$j = 1, \dots, r-1$$

is constant across parents. Since $Rd = 0$, child layer $j = 0$ is also constant across parents and the common child profile has zero sum. Hence

$$\ker \mathcal{O}_{r-1,r} = \mathcal{N}_r.$$

The recurrence shows that no later observation reduces this kernel, so

$$\ker \mathcal{O}_{L,r} = \mathcal{N}_r \qquad (L \geq r-1).$$

Therefore every realizable fiber is

$$x_\star + \mathcal{N}_r$$

and has dimension $r-1$. □

Corollary 1 (Sharp discrete minimal child-count law)**.** *Let $P \geq 2$ and $L \geq 0$. For a fixed number of update steps L, the minimal number of children per parent needed to realize every mass-conserving history in $\mathcal{M}_{P,L}$, over unrestricted real-valued fine states and for $0 < \lambda \leq 1$, is*

$$r_{\min} = L + 1.$$

Proof. By Theorem 1, $r = L + 1$ is sufficient.

Conversely, suppose $r \leq L$. Choose a nonzero vector

$$v \in \mathbb{R}_0^P.$$

Define a mass-conserving history through time L by

$$y^0 = \cdots = y^{r-1} = 0, \qquad y^r = v,$$

and choose any zero-sum states for later times, for example

$$y^{r+1} = \cdots = y^L = 0.$$

The recurrence (1) at $n = 0$ predicts

$$y^r = 0,$$

because

$$y^0 = \cdots = y^{r-1} = 0.$$

This contradicts $y^r = v \neq 0$. Hence the history is mass-conserving but not realizable. Therefore $r \leq L$ is insufficient. □

Example 2 (Nonnegative mass-conserving but nonrealizable history)**.** *Let*

$$P = 2, \qquad r = 2, \qquad L = 2, \qquad \lambda = \frac{1}{2}.$$

Consider

$$y^0 = (1, 1), \qquad y^1 = (1, 1), \qquad y^2 = (2, 0).$$

The history is nonnegative and conserves the sum of parent averages. However, the lag-two recurrence (1) *with*

$$y^0 = y^1 = (1, 1)$$

predicts

$$y^2 = (1, 1).$$

The recurrence residual is therefore

$$(1, -1),$$

so no fine initial state with two children per parent realizes this two-step parent history.

The minimal child-count law is a discrete-step law. If parent width is fixed and r changes, then fine-cell width changes. At fixed Courant number, the fine time step changes as well. Thus the law does not claim a fixed-physical-time mesh-resolution criterion.

5 Exact Computational Certificates

The accompanying script verifies representative finite instances of the theorems and generates the reported exact rational examples. The general results are established analytically above. All certificate calculations use Python's `fractions.Fraction`, so no floating-point roundoff enters the reported values.

Table 1: Exact rational certificates generated by the accompanying script. All checks are performed with Python's `fractions.Fraction`, so the results do not depend on floating-point roundoff. The first row constructs the normalized short-horizon lift in Example 1. The second row checks that a generated parent history satisfies the order-r recurrence. The third row gives the nonnegative nonrealizable mass-conserving history in Example 2.

Certificate	Parameters	Result
Normalized short-horizon lift	$P = 2,\ L = 1,\ r = 2,\ \lambda = 1/2$	exact lift
Long-horizon recurrence	$P = 3,\ r = 3,\ L = 6,\ \lambda = 1/2$	residual zero
Nonrealizable nonnegative history	$P = 2,\ r = 2,\ L = 2,\ \lambda = 1/2$	residual $(1, -1)$

Table 1 provides reproducibility checks for the finite examples. It is not used as a substitute for the analytical proofs.

6 Discussion

The present note is the image-space counterpart of the finite-horizon observation analysis in [10], which characterizes rank, nullspace, conditioning, and predictive memory of the forward history map. It is also complementary to the one-step information-boundary analysis in [11]. Here we solve the inverse problem: we characterize the image of the history map, construct a normalized right inverse, and parameterize the fine-state fibers over every realizable parent history.

The finite-volume-specific content is not merely that finite-dimensional linear trajectories satisfy a recurrence. The recurrence here has an explicit form induced by

$$S^r = \text{one parent-level shift},$$

begins sharply after the first r parent states, and is paired with a constructive fine-grid lifting map. The short-horizon fibers separate periodic flux-gauge freedom from interior child values that have not yet reached a parent interface.

This is not a validation theorem for nonlinear CFD. It is an exact algebraic statement for a declared periodic scalar upwind hierarchy over unrestricted real-valued fine states. Positivity, maximum principles, entropy admissibility, physical state constraints, nonlinear systems, nonperiodic boundaries, high-order reconstruction, and multistage integration require separate analysis.

7 Conclusion

For $P \geq 2$, $0 < \lambda \leq 1$, and unrestricted real-valued fine states, every mass-conserving parent history over L update steps in the periodic scalar upwind hierarchy is exactly liftable to a fine-grid initial state if and only if the number of children per parent satisfies

$$r \geq L + 1.$$

The lift can be chosen in a normalized canonical form by fixing the zero-mean periodic flux gauge, inverting the collar-to-flux map, setting unexposed interior children to zero, and using the $j = 0$ child to enforce the initial parent averages.

When $r < L + 1$, mass conservation is insufficient. The parent history must satisfy the explicit lag-r recurrence (1). The full fiber structure is also determined: short-horizon lift ambiguity consists of periodic flux gauges and unexposed interior children, whereas long-horizon lift ambiguity reduces to the repeated zero-mean parent profile.

The result provides a sharp discrete minimal child-count law and a constructive image-and-fiber characterization for the periodic upwind hierarchy. Together with the finite-horizon memory paper [10] and the one-step information-boundary note [11], this result provides a three-part algebraic baseline for the scalar periodic upwind hierarchy: recursive prediction requires exact memory, one-step closure requires the correct interface and entropy-side information, and coarse histories themselves require sufficient subcell resolution to be realizable.

Funding

No external funding was received for this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Antonis Polemitis: Conceptualization, Methodology, Verification, Writing–review and editing. Nicholas Christakis: Conceptualization, Methodology, Formal analysis, Software, Verification, Writing–original draft, Writing–review and editing. Dimitris Drikakis: Conceptualization, Methodology, Verification, Writing–review and editing.

Data and code availability

The complete reproducibility package for this study is publicly available in the Certified Simulation repository of the University of Nicosia, https://github.com/UniversityOfNicosia/certified-simulation, under `papers/coarse-history-realizability`. The package contains the scripts that generate the computational certificates, CSV files, and LaTeX table fragments reported in the manuscript, together with an independent certificate suite that replays the paper's exact results in exact rational arithmetic. The table entries and finite certificate checks are also generated by the accompanying Python scripts using exact rational arithmetic via `fractions.Fraction`. Thus, the reported finite identities, recurrence residuals, and table entries do not depend on floating-point roundoff. The scripts serve as reproducibility certificates for the finite examples; the mathematical claims are proved analytically above. No external CFD solver or proprietary software is required.

References

[1] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Cambridge Texts in Applied Mathematics. Cambridge University Press, Cambridge, 2002.

[2] Robert Eymard, Thierry Gallouët, and Raphaèle Herbin. Finite volume methods. In *Handbook of Numerical Analysis*, volume 7, pages 713–1018. Elsevier, 2000.

[3] Marsha J. Berger. On conservation at grid interfaces. *SIAM Journal on Numerical Analysis*, 24(5):967–984, 1987.

[4] Marsha J. Berger and Phillip Colella. Local adaptive mesh refinement for shock hydrodynamics. *Journal of Computational Physics*, 82(1):64–84, 1989.

[5] Peter McCorquodale and Phillip Colella. A high-order finite-volume method for conservation laws on locally refined grids. *Communications in Applied Mathematics and Computational Science*, 6(1):1–25, 2011.

[6] Stanley Osher and Richard Sanders. Numerical approximations to nonlinear conservation laws with locally varying time and space grids. *Mathematics of Computation*, 41(164):321–336, 1983.

[7] Emil M. Constantinescu and Adrian Sandu. Multirate timestepping methods for hyperbolic conservation laws. *Journal of Scientific Computing*, 33(3):239–278, 2007.

[8] Yohai Bar-Sinai, Stephan Hoyer, Jason Hickey, and Michael P. Brenner. Learning data-driven discretizations for partial differential equations. *Proceedings of the National Academy of Sciences of the United States of America*, 116(31):15344–15349, 2019.

[9] Dmitrii Kochkov, Jamie A. Smith, Ayya Alieva, Qing Wang, Michael P. Brenner, and Stephan Hoyer. Machine learning-accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences of the United States of America*, 118(21):e2101784118, 2021.

[10] Antonis Polemitis, Nicholas Christakis, and Dimitris Drikakis. Exact finite-horizon memory, conditioning, and dissipative decay in coarse upwind finite-volume prediction, 2026. arXiv:2608.08633.

[11] Antonis Polemitis, Nicholas Christakis, and Dimitris Drikakis. Initial-trace insufficiency and entropy-accounting separation in finite-volume coarse graining, 2026.

[12] Jan C. Willems. From time series to linear system—part II. exact modelling. *Automatica*, 22(6):675–694, 1986.

[13] Jan Willem Polderman and Jan C. Willems. *Introduction to Mathematical Systems Theory: A Behavioral Approach*, volume 26 of *Texts in Applied Mathematics*. Springer, New York, 1998.

[14] Ivan Markovsky and Florian Dörfler. Behavioral systems theory in data-driven analysis, signal processing, and control. *Annual Reviews in Control*, 52:42–64, 2021.

[15] Stephen E. Cohn and Dick P. Dee. Observability of discretized partial differential equations. *SIAM Journal on Numerical Analysis*, 25(3):586–617, 1988.

[16] B. L. Ho and Rudolf E. Kalman. Effective construction of linear state-variable models from input/output functions. *Regelungstechnik*, 14(12):545–548, 1966.

[17] Bruce C. Moore. Principal component analysis in linear systems: Controllability, observability, and model reduction. *IEEE Transactions on Automatic Control*, 26(1):17–32, 1981.

[18] G. David Forney, Jr. Minimal realizations of linear systems: The "shortest basis" approach. *IEEE Transactions on Information Theory*, 57(2):726–737, 2011.

[19] Athanasios C. Antoulas. *Approximation of Large-Scale Dynamical Systems*. Society for Industrial and Applied Mathematics, Philadelphia, 2005.

[20] Peter Benner, Serkan Gugercin, and Karen Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015.

[21] Benjamin Peherstorfer and Karen Willcox. Data-driven operator inference for nonintrusive projection-based model reduction. *Computer Methods in Applied Mechanics and Engineering*, 306:196–215, 2016.

[22] Ioannis G. Kevrekidis, C. William Gear, James M. Hyman, Panagiotis G. Kevrekidis, Olof Runborg, and Constantinos Theodoropoulos. Equation-free, coarse-grained multiscale computation: Enabling microscopic simulators to perform system-level analysis. *Communications in Mathematical Sciences*, 1(4):715–762, 2003.